

Shahir Abdullah

Software Engineer

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EDUCATION

Exam	Institution	Major	Duration	Result
M.Sc.	United International University	Computer Science & Engineering	2026 – Present	Ongoing
B.Sc.	Military Institute of Science and Technology	Computer Science & Engineering	2017 – 2021	CGPA 3.21 / 4.00
				Final Year: 3.73 / 4.00
HSC	Mirzapur Cadet College	Science	2016	GPA 5.00 / 5.00
SSC	Mirzapur Cadet College	Science	2014	GPA 5.00 / 5.00

CERTIFICATIONS

Course	Link
Coursera Basic Generative Adversarial Networks	View
Coursera Advanced Generative Adversarial Networks	View
Coursera Fundamentals of Reinforcement Learning	View

PUBLICATIONS

0.1 A Point-of-Prescription Safety-Check System for Adverse Drug Reactions in Rural Bangladeshi Hospitals: A Feasibility Study

Shahir Abdullah

(In preparation)

Proposes a lightweight, smartphone-based decision-support system that surfaces a patient's severe adverse-drug-reaction history at the point of prescription in rural Bangladeshi hospitals, where no outpatient record system exists and physicians have under a minute per patient. The system captures a soft patient identifier at registration, reads the prescription, resolves brand names to active ingredients via national drug references, and checks them against the patient's recorded severe-reaction history using retrieval rather than prediction, staying silent unless a high-risk match is found. Evaluated as a feasibility study assessing workflow fit, usability, and retrospective detection of known reaction cases.

Links: [arXiv](#) — [Google Scholar](#)

0.2 Diagonal Scaling : A Multi-Dimensional Resource Model and Optimization Framework for Distributed Databases

Shahir Abdullah, [Syed Rohit Zaman]

(In preparation)

Proposes a formal model for *diagonal scaling* — a hybrid approach that dynamically balances vertical and horizontal scaling to optimize cost, performance, and reliability in distributed systems. The work introduces a scaling plane abstraction, cost-performance trade-off functions, and a control algorithm for adaptive resource allocation under variable load.

Links: [arXiv](#) — [Google Scholar](#)

RESEARCH INTEREST

LLM, Autonomous Intelligent systems, HCI, Health Informatics, AI, Cloud Computing, Distributed Systems, Reinforcement learning

RESEARCH EXPERIENCE

Diagonal Scaling: A Multi-Dimensional Resource Model and Optimization Framework for Distributed Databases (Ongoing)

Developing a formal framework for *diagonal scaling* that models vertical and horizontal resource allocation as a multi-dimensional optimization problem. The work defines a scaling plane, cost-performance trade-off functions, and adaptive control strategies to optimize latency, throughput, and cost in distributed database systems under dynamic workloads.

A Point-of-Prescription Safety-Check System for Adverse Drug Reactions in Rural Bangladeshi Hospitals: A Feasibility Study (Ongoing)

Developing a lightweight, smartphone-based clinical decision-support system that surfaces a patient's severe adverse-drug-reaction history at the point of prescription in rural Bangladeshi hospitals, where no outpatient record system exists and a physician may

have under a minute per patient.

UNDERGRADUATE RESEARCH AND PROJECTS

App-Based Bangla Book Reader System for the Blind

Capstone Project

Developed an end-to-end assistive system that converts printed Bangla text into speech for visually impaired users.

- Built an Android application that captures images of book pages and sends them to a backend server for processing.
- Implemented an OCR and text recognition pipeline using **OpenCV EAST Text Detector** for word segmentation and a **CNN (EfficientNet-B0)** model for character recognition, achieving **85% accuracy**.
- Integrated Google's **Bangla Text-to-Speech API** to convert recognized text into natural-sounding audio output.
- Designed the hardware setup using a custom book stand, smartphone, Arduino, physical buttons, and a Bluetooth module to improve usability for blind users.

Links: [YouTube Demo](#) — [Backend Code](#) — [Android Code](#)

Handwritten Geometric Shape Detector

Machine Learning Project

Developed a desktop application that recognizes hand-drawn geometric shapes using a convolutional neural network.

- Built a Python desktop application that allows users to draw geometric shapes and classifies them in real time.
- Trained a **CNN** on a dataset of over **2,000** handwritten images of circles, squares, rectangles, and triangles.
- Implemented prediction confidence scoring to display model certainty alongside classification results.

Links: [YouTube Demo](#) — [Code](#)

Digital Image Processing Library

Image Processing Project

Created a modular Python library implementing fundamental image processing algorithms for educational and experimental use.

- Implemented common filters and transformations including convolution, edge detection, smoothing, and thresholding.
- Provided interactive examples and documentation using **Jupyter Notebook** for easy experimentation and learning.

Links: [Code](#)

E-Waste Management System

Database Management Systems Course Project

Designed and implemented a database-driven web system to manage the collection, tracking, and processing of electronic waste.

- Designed a normalized relational database schema to model users, collection requests, waste categories, logistics, and processing centers.
- Implemented CRUD operations and transactional workflows using **MySQL** and a web-based application interface.
- Focused on data integrity, query optimization, and consistency for reliable record management.

PROFESSIONAL EXPERIENCE

Software Engineer 2 — Ajentica

Jun 2026 – Present

- Architected a multi-layer Windows UI Automation capture engine in **C#/.NET 8** for Lumistry, a PMS company, instrumenting **25+ distinct event** types across **3 independent OS-level subsystems** (UIA automation/property events, Win32 WinEvent/MSAA hooks, and a global low-level mouse hook) to extract structured interaction data from a legacy WinForms application with zero native automation support.
- Built an async, lock-free event pipeline (**BlockingCollection producer/consumer + dedicated drain thread**) to serialize events to JSONL, decoupling latency-sensitive UI-thread callbacks from disk I/O and keeping hook response times non-blocking.
- Automated the full release process with a 2-stage GitLab **CI/CD pipeline** that cross-compiles a self-contained single-file win-x64 executable on Linux runners and packages it into a versioned NSIS installer — reducing release packaging to a single tagged commit.
- Took ownership and communicated with Client to successfully deliver the POC while applying AI-DLC to streamline delivery, reducing average feature time-to-market by **80%** through synchronized release planning.
- Participated in POC, LLM agentic deliverable architecture reviews to optimize designs, reducing development time by **25%**, lowering the time to deliver by **10%**.
- Troubleshoot production issues and customer-reported defects, reducing average resolution time while maintaining high service reliability.
- Leverage AI-assisted development tools including **GitHub Copilot** and **Anthropic Claude** to accelerate implementation, documentation, debugging, and code review workflows by applying AI-DLC.

- Designed and developed enterprise-scale software solutions using **C#**, **ASP.NET Core**, **Angular**, **SQL Server** and **Postgres** for **IQVIA**, a Fortune 500 healthcare technology company serving **10,000+** healthcare and life sciences organizations across **100+** countries.
- Built and maintained reusable APIs, modular frontend components, and shared application services that reduced feature development effort by **10%** and improved maintainability across multiple product teams.
- Optimized data-intensive workflows handling datasets containing **10K+** records, reducing page load times by **50%** and backend processing times by **10%**.
- Collaborated across **5+** cross-functional teams (PMs, Architects, Designers, QA, and DevOps) to streamline the SDLC, reducing average feature time-to-market by **20%** through synchronized release planning.
- Participated in backend architecture reviews to optimize designs, reducing development time by **15%**, lowering AWS cloud infrastructure costs by **5%**.
- Troubleshoot production issues and customer-reported defects, reducing average resolution time by **15%** while maintaining high service reliability.
- Develop automated unit and integration tests, increasing code coverage from **80%** to **96%** and improving release confidence.
- Leverage AI-assisted development tools including GitHub Copilot and Cursor to accelerate implementation, documentation, debugging, and code review workflows.

Junior Full-Stack Engineer — BiTechX

Dec 2019 – Mar 2020

- Developed web applications using **Laravel** following MVC architecture principles.
- Designed responsive UI components with **JavaScript**, **HTML**, and **CSS**.
- Designed relational database schemas and managed application data using **MySQL**.
- Built and tested RESTful APIs and performed integration testing with **Postman**.

AI Engineer Trainee — Intelligent Machines Ltd

Jan 2020 – Apr 2020

- Worked on computer vision and image processing pipelines using **Python**.
- Assisted in model design, optimization, and fine-tuning for machine learning applications.
- Conducted experiments to evaluate model performance and improve inference efficiency.

SKILLS

- **Languages:** C#, TypeScript, Python, JavaScript, SQL, Bash
- **Frameworks:** .NET Core, Angular, FastAPI, Laravel, TensorFlow, OpenCV
- **Databases:** MySQL, PostgreSQL, SQL Server
- **Cloud & DevOps:** DigitalOcean, Apache, Nginx, Docker (basic), systemd, CI/CD
- **Frontend:** Angular, RxJS, SCSS, Responsive UI, State Management
- **ML & AI:** CNNs, EfficientNet, OCR, Image Processing
- **Engineering:** Distributed Systems, Async Systems, SOLID, BDD
- **Tools:** Git, GitHub, GitLab, Postman, Linux, Jupyter

REFERENCES

- **Dr. Muhammad Nazrul Islam** — Associate Professor, CSE, MIST
Email: nazrul@cse.mist.ac.bd — [Google scholar link](#)
- **Dr. T. M. Shahriar Sazzad** — Associate Professor, CSE, MIST
Email: shahriar@cse.mist.ac.bd — [Google scholar link](#)
- **Seeam Shahid Noor** — Product Manager, Scale AI — Co founder, Nour Technologies
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